MVP Simulator: MDP Formulation and Initial Results

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1 MDP Formulation

The MVP simulator implements a finite-horizon MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma, T)$:

1.1 State Space

$$\mathcal{S} = \{\texttt{sedentary},\ \texttt{active}\}$$

Two activity states with no additional time-of-day dimension. Time-dependent reward scaling is handled via `reward_multiplier_by_step` (see §1.4 below).

1.2 Action Space

$$\mathcal{A} = \{\texttt{nudge},\ \texttt{idle}\}$$

A binary intervention: send a motivational message (`nudge`) or do not (`idle`).

1.3 Transition Function

$P(s' \mid s, a)$ is a configurable $2 \times 2 \times 2$ tensor. Rows index current state $s \in \{\text{sedentary}, \text{active}\}$ and columns index next state s' in the same order:

$$P_{\text{nudge}} = \begin{bmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{bmatrix} \qquad P_{\text{idle}} = \begin{bmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{bmatrix}$$

Sending a reminder increases $P(\text{active} \mid \text{sedentary})$ from 0.1 to 0.3 but decreases $P(\text{active} \mid \text{active})$ from 0.6 to 0.5, consistent with Reactance theory (reminders can backfire for already-active users).

Optional `reward_multiplier_by_step`: a per-step scaling factor that can zero out rewards at certain daily steps (soft mask). When absent, all multipliers default to 1.0. The per-step reward is computed as base reward $\times$ multiplier$[t]$ where t is the step-of-day index.

1.4 Reward

The base reward is a vector indexed by next state:

$$r = [0.0,\ 1.0]^\top \quad (\text{sedentary},\ \text{active})$$

The per-step reward at time t scales by the multiplier:

$$r(s, a, t) = m[t \bmod S] \cdot r(s')$$

where $S =$ `steps_per_day`, $m \in \mathbb{R}^S$ is `reward_multiplier_by_step` (default $m_i = 1.0$), and s' is the next state after taking action a in state s. Currently $m_i = 1.0$ for all i, so $r(s, a, t) = r(s')$.

Single-timescale placeholder. Multi-timescale reward (immediate + delayed body measure) is deferred to Phase 2.

1.5 Episode Structure

$T =$ `episode_days` $\times$ `steps_per_day` timesteps. Default: $T = 90$ days $\times$ 5 steps/day $= 450$.

2 Agents

All agents are contextual bandits: the state is observed but ignored during action selection. Because the transition matrix is stationary, each action a induces a Markov chain whose stationary distribution π_a follows from P_a:

$$\pi_a(\text{active}) = \frac{P(\text{active} \mid \text{sedentary}, a)}{P(\text{active} \mid \text{sedentary}, a) + P(\text{sedentary} \mid \text{active}, a)}, \qquad \pi_a(\text{sedentary}) = 1 - \pi_a(\text{active})$$

The expected reward for always choosing action a is:

$$E[r \mid a] = \sum_{s'} \pi_a(s')\, r(s') = \pi_a(\text{active})$$

From the MVP matrices:

$$E[r \mid \text{nudge}] = \frac{0.3}{0.3 + 0.5} \approx 0.375, \qquad E[r \mid \text{idle}] = \frac{0.1}{0.1 + 0.4} = 0.20$$

Nudge dominates idle in expectation, but a state-aware agent could improve further by conditioning on s (nudge when sedentary, idle when active), yielding $\pi(\text{active}) \approx 0.429$. This limitation motivates the state-aware agents planned for Phase 2.

2.1 Thompson Sampling

Beta-Bernoulli TS with prior Beta$(1, 1)$ per action. Posterior update:

$$\alpha_a \leftarrow \alpha_a + \mathbb{1}[s'.\texttt{activity} = \texttt{active}], \quad \beta_a \leftarrow \beta_a + \mathbb{1}[s'.\texttt{activity} = \texttt{sedentary}]$$

Action selected by sampling $\theta_a \sim \text{Beta}(\alpha_a, \beta_a)$ and choosing $a^* = \arg\max_a \theta_a$.

2.2 Epsilon-Greedy

$\epsilon = 0.1$. With probability ϵ, select uniformly at random; otherwise select $\arg\max_a Q(a)$. Q-values updated via incremental average:

$$Q(a) \leftarrow Q(a) + \frac{r_t - Q(a)}{n_a}$$

2.3 UCB1

Upper Confidence Bound with exploration parameter $c = 2.0$. Selects:

$$a^* = \arg\max_a \left[Q(a) + c\sqrt{\frac{\ln N}{N_a}} \right]$$

where N is total steps and N_a is the number of times action a was chosen. The confidence bonus shrinks as N_a grows, naturally shifting from exploration to exploitation. Each action is pulled once before UCB is applied.

2.4 Random Baseline

Uniform random action selection. No learning. Serves as lower bound.

3 Initial Results

All agents run over 10 random seeds (1–10), 450 timesteps each. Results are reported as mean ± standard deviation across seeds.

3.1 Standard Config (`config/rule_based.yaml`)

No `reward_multiplier_by_step` – uniform 1.0 multiplier on all steps.

Agent	Total Reward	Mean Reward/Step
Thompson Sampling ($\alpha_0 = \beta_0 = 1$)	160.3 ± 10.5	0.356 ± 0.023
Epsilon-Greedy ($\epsilon = 0.1$)	160.8 ± 15.3	0.357 ± 0.034
UCB ($c = 2.0$)	148.2 ± 15.0	0.329 ± 0.033
Random	135.6 ± 10.7	0.301 ± 0.024

Table 1: Results on `config/rule_based.yaml` (10 seeds, 450 timesteps). TS and ϵ-greedy perform similarly; TS has lower variance. UCB outperforms random but shows higher variance.

3.2 Masked Config (`config/rule_based_with_mask.yaml`)

`reward_multiplier_by_step: [1, 1, 1, 1, 0]` – step 4 (end of day) yields zero reward regardless of outcome. All agents drop proportionally to the masked fraction (1/5 of steps zeroed), confirming the multiplicative reward model.

Agent	Total Reward	Mean Reward/Step
Thompson Sampling ($\alpha_0 = \beta_0 = 1$)	126.6 ± 6.8	0.281 ± 0.015
Epsilon-Greedy ($\epsilon = 0.1$)	127.3 ± 10.2	0.283 ± 0.023
UCB ($c = 2.0$)	116.9 ± 8.6	0.260 ± 0.019
Random	107.8 ± 6.7	0.240 ± 0.015

Table 2: Results on `config/rule_based_with_mask.yaml` (10 seeds, 450 timesteps). Reward drops $\approx 20\%$ across all agents, matching the 1/5 masked fraction. Relative ordering is preserved.

4 Known Limitations

- Agents are state-blind bandits, not state-aware RL. State-aware agents are Phase 2.
- Single-timescale reward only. Multi-timescale (immediate + 3-week delayed) is Phase 2.
- No user response model. Burden accumulation and 4 archetypes are Phase 2.
- Transition matrix is fixed and known. Learned transitions from real data are Phase 2.
- Transition probabilities are placeholder values informed by Reactance theory. Calibration against real data is Phase 2.